

A volume-grid-free history-integral model for wake-memory dynamics of patterned autochemotactic particles

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Autochemotactic particles respond to chemical fields deposited along their own trajectories, which gives their motion a nonlocal memory of past positions. We study this coupling for particles with surface-patterned activity and response. The chemical field obeys free-space diffusion with moving point or permeable surface sources. Its heat-kernel representation replaces the bulk field solve by an integral over the source history, while retaining a discretization of the particle surface. We split the temporal convolution into a smooth history contribution and a near-present local contribution. The local calculation uses a one-step trajectory approximation, an inverse-time transformation, and a regularized treatment of surface self-interaction. This construction extends the point-source Green-function formulation used in our previous work to rotating surface distributions and chemically induced translation and rotation. Representative isotropic trajectories agree in morphology with calculations from the earlier grid implementation. For two surface patterns, the diffusion-dominated limit gives Fourier-series reference solutions specific to a permeable circle. Two patterned-particle configurations produce localized loops and, in some reported cases, elongated, slingshot-like trajectories. Concentration overlays associate these changes with delayed encounters between the particle and its chemical wake. The model omits ambient flow and fluid-mediated solute advection; its predictions concern a reduced, deterministic description of chemical-memory-driven motion.

I. INTRODUCTION

Autochemotactic particles move in fields that record their previous positions. Diffusion broadens this chemical record, while surface activity and chemical response determine how a particle reacts when it encounters the field again. The resulting feedback couples translation, rotation, and trajectory history in biological and synthetic active matter.^{1–4} Experiments on active droplets and Janus particles provide examples in which chemical production, solubilization, or fuel consumption accompanies self-propulsion.^{5–8} Chiral droplets add an intrinsic or emergent rotation and can produce curling or helical trajectories.^{9,10}

Particle-based models isolate chemical memory from the interfacial and hydrodynamic processes that generate propulsion. In this description, a moving source produces a diffusing scalar field, and the particle responds to the field gradient. The model does not resolve solvent flow, interfacial stress, or advection by a phoretic flow. It remains useful when the goal is to test whether delayed chemical feedback can generate a given trajectory within

a controlled reduced model.

Eulerian discretizations solve the chemical field throughout the surrounding volume. For free-space diffusion with prescribed moving sources, the same field can be written as a convolution of the heat kernel with the source history. This representation evaluates the concentration or its gradient only where the particle dynamics require it. A finite surface source introduces a sharply localized near-present self-interaction and a source–target surface sum, so the volume-grid-free representation still requires quadrature and regularization.

A trajectory-based Green-function representation appeared in our previous study of isotropic chiral droplets.¹⁰ Appendix E of that work gave the point-source history integral and compared Green-function and grid-based trajectories. We use that formulation as the starting point. The Green-function representation itself is not a new contribution of the present paper. We extend it to rotating, surface-distributed activity, introduce a history–local treatment of the most recent source history, and couple the computed surface gradients to translational and rotational response patterns.

The paper has one numerical extension and one physical application. We first define a deterministic, chemically permeable moving-source model and its regime of validity. We then formulate a direct history integral for surface sources and state how the local self-

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interaction is regularized. Analytical calculations for patterned circular particles provide diffusion-dominated reference trends, while representative trajectories cross-check the implementation against the earlier grid calculations. We finally use two activity–response patterns to examine how a particle can re-encounter its deposited chemical wake. The computed circular and slingshot-like trajectories support a wake-memory interpretation within the reduced model; they do not constitute predictions for a fully coupled hydrodynamic droplet.

II. KINEMATIC AUTOCHEMOTACTIC MODEL

A. Scope and assumptions

We consider a chemically active circle in two dimensions or a sphere in three dimensions. The particle has radius R , center $\mathbf{x}_p^*(t^*)$, and body orientation $\mathbf{Q}(t^*)$. Here $\mathbf{Q}$ denotes a planar rotation in two dimensions and an element of $\text{SO}(3)$ in three dimensions. A body-fixed point $\boldsymbol{\xi} \in \partial B_1$ lies at

$$\mathbf{y}^*(\boldsymbol{\xi}, t^*) = \mathbf{x}_p^*(t^*) + R\mathbf{Q}(t^*)\boldsymbol{\xi}. \quad (1)$$

The body-fixed activity $a^*(\boldsymbol{\xi})$ is positive for production and negative for consumption.

The scalar diffusivity D is constant throughout the domain and has the same value inside and outside the particle. The scalar therefore crosses the nominal particle boundary. This assumption represents a permeable moving source surface in a homogeneous medium rather than an impermeable solid particle. We write $c^*(\mathbf{x}^*, t^*)$ for the deviation from a uniform background concentration. It satisfies

$$\begin{aligned} \frac{\partial c^*}{\partial t^*} &= D\nabla^{*2}c^* + S^*(\mathbf{x}^*, t^*), \\ S^*(\mathbf{x}^*, t^*) &= R^{d-1} \int_{\partial B_1} a^*(\boldsymbol{\xi}) \delta_d(\mathbf{x}^* - \mathbf{x}_p^* - R\mathbf{Q}\boldsymbol{\xi}) dS_{\boldsymbol{\xi}}. \end{aligned} \quad (2)$$

We impose

$$c^*(\mathbf{x}^*, 0) = 0, \quad c^*(\mathbf{x}^*, t^*) \rightarrow 0 \quad \text{as } |\mathbf{x}^*| \rightarrow \infty \quad (4)$$

for each finite t^* . The surface-source representation gives continuity of c^* and a normal-flux jump proportional to a^* . In two dimensions, a nonzero net source has no bounded steady concentration in an unbounded domain. The finite-time solution and its gradient remain well defined. Steady calculations below use the mean-free field, because the spatially uniform angular mode does not affect the tangential gradient.

For an isotropic particle, the source can be replaced by a point source when finite-size surface structure is irrelevant. A source of total rate Q_0^* then has

$$S^*(\mathbf{x}^*, t^*) = Q_0^* \delta_d(\mathbf{x}^* - \mathbf{x}_p^*(t^*)). \quad (5)$$

B. Kinematic surface response

The chemical field affects the particle through a phenomenological kinematic closure motivated by phoretic response. We define the translational drift and angular velocity by

$$\mathbf{U}_c^* = -\frac{1}{A_R} \int_{\partial S(t)} M_t^*(\mathbf{s}) \nabla_s^* c^* dS, \quad (6)$$

$$\boldsymbol{\omega}_c^* = -\frac{1}{A_R R} \int_{\partial S(t)} M_r^*(\mathbf{s}) \mathbf{n} \times \nabla_s^* c^* dS, \quad (7)$$

where $A_R = A_d R^{d-1}$, $A_2 = 2\pi$, and $A_3 = 4\pi$. The functions M_t^* and M_r^* are effective translational and rotational surface-response coefficients. Equations (6) and (7) define the reduced model. The resulting quantities are drift and angular velocity, not mechanical force and torque. A hydrodynamic phoretic model would instead derive rigid-body motion from a slip condition and a coupled Stokes–solute boundary-value problem.^{11,12}

The particle motion obeys

$$\frac{d\mathbf{x}_p^*}{dt^*} = v_0^* \mathbf{p} + \mathbf{U}_c^*, \quad (8)$$

$$\frac{d\mathbf{Q}}{dt^*} = [\boldsymbol{\omega}_0^* + \boldsymbol{\omega}_c^*] \times \mathbf{Q}, \quad (9)$$

with prescribed initial position and orientation. In two dimensions, Eq. (9) reduces to an equation for one orientation angle. The activity and response patterns rotate with $\mathbf{Q}(t^*)$.

C. Dimensionless formulation

Let $a_* > 0$ and $M_* > 0$ denote fixed input scales for activity and response. We choose a reference speed $U_* > 0$ before a simulation. For an intrinsically propelled particle, one may set $U_* = v_0^*$. For a particle without intrinsic propulsion, one possible input-based choice is $U_* = M_* a_*/D$. Whichever input scale is selected, it is held fixed over a parameter sweep and is not the computed terminal speed.

We introduce

$$\mathbf{x} = \frac{\mathbf{x}^*}{R}, \quad t = \frac{U_* t^*}{R}, \quad C = \frac{U_* c^*}{a_*}, \quad (10)$$

and define

$$\text{Pe}_{\text{in}} = \frac{RU_*}{D}, \quad \alpha = \frac{v_0^*}{U_*}, \quad \boldsymbol{\Omega}_0 = \frac{R\boldsymbol{\omega}_0^*}{U_*}, \quad (11)$$

$$\sigma(\boldsymbol{\xi}) = \frac{a^*(\boldsymbol{\xi})}{a_*}, \quad \Lambda_t(\boldsymbol{\xi}) = \frac{M_t^*(\boldsymbol{\xi})a_*}{U_*^2 R}, \quad \Lambda_r(\boldsymbol{\xi}) = \frac{M_r^*(\boldsymbol{\xi})a_*}{U_*^2 R}. \quad (12)$$

The subscript “in” distinguishes the prescribed diffusion parameter from the output swimming Péclet number introduced below. The dimensionless system is

$$\frac{d\mathbf{x}_p}{dt} = \alpha\mathbf{p} + \mathbf{U}_c, \quad (13)$$

$$\frac{d\mathbf{Q}}{dt} = [\mathbf{\Omega}_0 + \mathbf{\Omega}_c] \times \mathbf{Q}, \quad (14)$$

$$\frac{\partial C}{\partial t} = \frac{1}{\text{Pe}_{\text{in}}} \nabla^2 C + \int_{\partial B_1} \sigma(\boldsymbol{\xi}) \delta_d(\mathbf{x} - \mathbf{x}_p - \mathbf{Q}\boldsymbol{\xi}) dS_{\boldsymbol{\xi}}, \quad (15)$$

where

$$\mathbf{U}_c = -\frac{1}{A_d} \int_{\partial B_1} \Lambda_t(\boldsymbol{\xi}) \nabla_s C(\mathbf{x}_p + \mathbf{Q}\boldsymbol{\xi}, t) dS_{\boldsymbol{\xi}}, \quad (16)$$

$$\mathbf{\Omega}_c = -\frac{1}{A_d} \int_{\partial B_1} \Lambda_r(\boldsymbol{\xi}) (\mathbf{Q}\boldsymbol{\xi}) \times \nabla_s C(\mathbf{x}_p + \mathbf{Q}\boldsymbol{\xi}, t) dS_{\boldsymbol{\xi}}. \quad (17)$$

For a computed dimensional speed U_{swim}^* , the output quantity is

$$\text{Pe}_{\text{swim}} = \frac{RU_{\text{swim}}^*}{D} = \text{Pe}_{\text{in}} \frac{U_{\text{swim}}^*}{U_*}. \quad (18)$$

For a stationary pattern in the diffusion-dominated limit, write $C = \text{Pe}_{\text{in}} \bar{C} + o(\text{Pe}_{\text{in}})$. The leading mean-free field satisfies

$$\nabla^2 \bar{C} = - \int_{\partial B_1} \sigma(\boldsymbol{\xi}) \delta_d(\mathbf{x} - \boldsymbol{\xi}) dS_{\boldsymbol{\xi}}. \quad (19)$$

The corresponding drift contains the compound response $\text{Pe}_{\text{in}} \Lambda_t$. This factor must be retained when velocities derived from $\bar{C}$ are compared with simulations.

III. VOLUME-GRID-FREE HISTORY-INTEGRAL FORMULATION

A. Heat-kernel representation

For the free-space initial-value problem in Eq. (15), the heat kernel is

$$G(\mathbf{z}, \tau) = \left(\frac{\text{Pe}_{\text{in}}}{4\pi\tau} \right)^{d/2} \exp\left(-\frac{\text{Pe}_{\text{in}}|\mathbf{z}|^2}{4\tau} \right), \quad \tau > 0. \quad (20)$$

The surface-source solution is

$$C(\mathbf{x}, t) = \int_0^t \int_{\partial B_1} G(\mathbf{x} - \mathbf{x}_p(t') - \mathbf{Q}(t')\boldsymbol{\xi}, t - t') \times \sigma(\boldsymbol{\xi}) dS_{\boldsymbol{\xi}} dt'. \quad (21)$$

The gradient required by the kinematic closure follows from

$$\nabla G(\mathbf{z}, \tau) = -\frac{\text{Pe}_{\text{in}}}{2\tau} \mathbf{z} G(\mathbf{z}, \tau). \quad (22)$$

Equations (21) and (22) replace the volume discretization by a temporal history and, for a finite source, a surface integral. They apply to the free-space diffusion model defined in Sec. II; other boundaries require a different Green function or an additional boundary representation.

At a current surface target

$$\mathbf{x}_j(t) = \mathbf{x}_p(t) + \mathbf{Q}(t)\boldsymbol{\xi}_j, \quad (23)$$

we project the full gradient onto the tangent plane,

$$\nabla_s C_j = (\mathbf{I} - \mathbf{n}_j \mathbf{n}_j) \nabla C(\mathbf{x}_j, t). \quad (24)$$

For surface nodes with quadrature weights w_j , the discrete drift and angular velocity are

$$\mathbf{U}_c \approx -\frac{1}{A_d} \sum_{j=1}^{N_s} w_j \Lambda_{t,j} \nabla_s C_j, \quad (25)$$

$$\mathbf{\Omega}_c \approx -\frac{1}{A_d} \sum_{j=1}^{N_s} w_j \Lambda_{r,j} (\mathbf{Q}\boldsymbol{\xi}_j) \times \nabla_s C_j. \quad (26)$$

Equal weights may be used for a uniformly spaced circular boundary. Three-dimensional surface quadratures require their actual weights.

B. History-local split

The heat kernel becomes localized near the current time. We split the gradient at a window length δ ,

$$\nabla C = \nabla C_H + \nabla C_L, \quad \delta = \Delta t, \quad (27)$$

where

$$\nabla C_H = \int_0^{t-\delta} \int_{\partial B_1} \nabla G \sigma dS_{\boldsymbol{\xi}} dt', \quad (28)$$

$$\nabla C_L = \int_{t-\delta}^t \int_{\partial B_1} \nabla G \sigma dS_{\boldsymbol{\xi}} dt'. \quad (29)$$

The arguments of G are those in Eq. (21). The history integrand has a positive separation in time and is evaluated with a composite trapezoidal rule over stored particle positions and orientations.

C. Regularized local contribution

Set $\tau = t - t'$. For a source–target pair with nonzero instantaneous separation, the exponential in Eq. (20) suppresses the kernel as $\tau \rightarrow 0$, although the peak can remain difficult to resolve with uniform quadrature. A diagonal difficulty occurs when a surface source approaches the same node at which the gradient is evaluated.

Over one local interval, we approximate the source point by

$$\mathbf{y}_k(t - \tau) \approx \mathbf{y}_k(t) - \mathbf{V}_k^* \tau, \quad (30)$$

where $\mathbf{V}_k^*$ is a one-sided velocity predictor available before the current explicit update. It includes translational motion and, when the surface pattern rotates over the local interval, the contribution $\mathbf{\Omega}^* \times (\mathbf{Q}\xi_k)$. The corresponding source–target vector is

$$\mathbf{R}_{jk}(\tau) \approx \mathbf{R}_{0,jk} + \mathbf{V}_k^* \tau, \quad \mathbf{R}_{0,jk} = \mathbf{x}_j(t) - \mathbf{y}_k(t). \quad (31)$$

For the surface-source calculation, we replace $|\mathbf{R}_{jk}(\tau)|^2$ in the exponential by $|\mathbf{R}_{jk}(\tau)|^2 + \epsilon^2$. The parameter ϵ defines a regularized numerical approximation to the surface self-interaction. We do not assign an a priori $O(\epsilon^2)$ error because an expansion in ϵ^2/τ is not uniform near $\tau = 0$.

After the change of variables $u = 1/\tau$, a regularized pair integral has the form

$$\mathbf{I}_{jk}^\epsilon = \int_{1/\delta}^\infty \mathbf{g}_{jk}^\epsilon(1/u) \frac{du}{u^2}. \quad (32)$$

Define

$$A_{jk}^\epsilon = \frac{\text{Pe}_{\text{in}}}{4} (|\mathbf{R}_{0,jk}|^2 + \epsilon^2). \quad (33)$$

For $A_{jk}^\epsilon > 0$, the substitution $x = A_{jk}^\epsilon(u - 1/\delta)$ isolates an e^{-x} weight,

$$\mathbf{I}_{jk}^\epsilon = \frac{e^{-A_{jk}^\epsilon/\delta}}{A_{jk}^\epsilon} \int_0^\infty e^{-x} \mathbf{h}_{jk}^\epsilon(x) dx. \quad (34)$$

We approximate Eq. (34) with M Gauss–Laguerre nodes. This transformation supplies a decaying weight for off-diagonal pairs and for regularized diagonal pairs. It does not define the unregularized $A_{jk} = 0$ boundary limit.

The local gradient at target j is assembled as

$$\nabla C_L(\mathbf{x}_j, t) \approx \sum_{k=1}^{N_s} w_k \sigma_k \sum_{m=1}^M \hat{w}_m \mathbf{f}_{jkm}^\epsilon, \quad (35)$$

where $\hat{w}_m$ includes the scaling in Eq. (34). The parameters M , ϵ , and N_s control distinct approximations.

D. Time update and computational cost

We use an explicit first-order update for the particle position and orientation. In two dimensions, updating the orientation angle preserves the unit orientation vector. A three-dimensional implementation should update $\mathbf{Q}$ with a rotation map rather than applying an unconstrained Euler step to a unit vector.

Let n denote the number of stored time levels at the current step. For a point source evaluated at N_s target nodes, direct history summation costs $O(nN_s)$ per step. A surface source with N_s source patches and N_s targets costs $O(nN_s^2)$ per step. The M -point local calculation adds $O(MN_s)$ and $O(MN_s^2)$ work, respectively. A direct N_t -step calculation therefore costs $O(N_t^2 N_s)$ for a point source and $O(N_t^2 N_s^2)$ for a surface source.

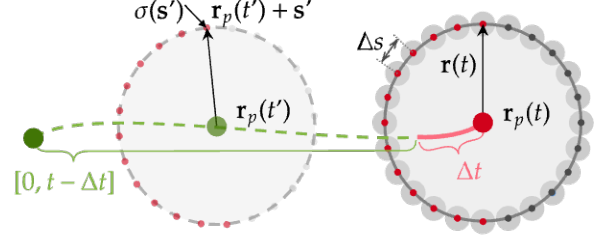


FIG. 1. **History–local decomposition.** The time convolution is split into a stored history over $[0, t - \Delta t]$ and a local contribution over the most recent step. The surrounding volume is not discretized. A surface source and the surface targets are represented by N_s quadrature nodes.

For N_p interacting surface-source particles, direct source–target summation costs $O(N_p^2 n N_s^2)$ at step n . Spatial acceleration could reduce the source–target work at a fixed history level, but it would not remove the quadratic accumulation over time. We have not implemented or benchmarked such acceleration. The present method removes the volume grid; it does not establish a computational speed advantage.

E. Accuracy assessment

The calculation contains four independent numerical parameters: Δt , N_s , M , and ϵ . A refinement that changes several parameters at once cannot identify their separate errors. For fixed (N_s, M, ϵ) , forward Euler has a formal first-order consistency expectation when the regularized history-dependent right-hand side is smooth and Lipschitz continuous. Discontinuous activity patterns and near-diagonal interactions can violate the regularity behind this statement.

The local trajectory approximation is also first order. Its error depends on the algebraic heat-kernel prefactor, body rotation, surface quadrature, inverse-time quadrature, and regularization. Perturbing only the exponential factor does not give a complete error estimate. We therefore make no theorem-level claim for the global convergence order of the surface-source algorithm.

IV. RESULTS

A. Cross-check for an isotropic point-source model

We first consider the point-source specialization of Eq. (15). The parameters and initial conditions are those used for the isotropic chiral-particle calculations in Ref.¹⁰. Figure 2 compares two trajectories from the present history-integral implementation with the corresponding grid-based calculations. Both implementations

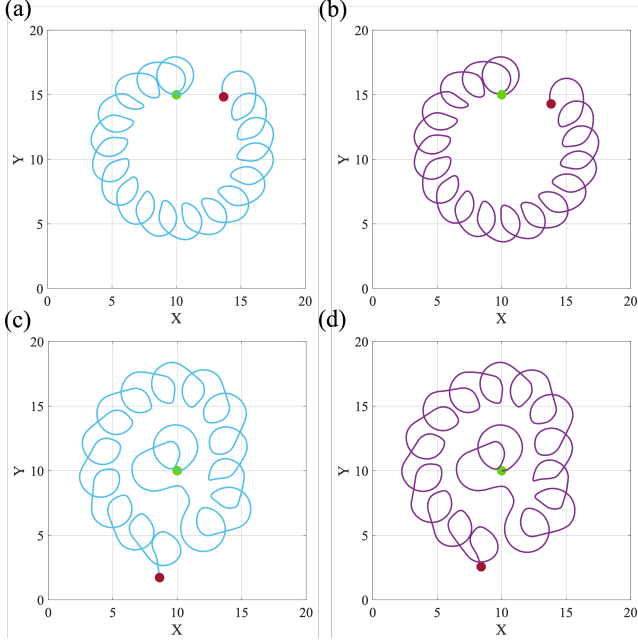


FIG. 2. **Qualitative comparison of curling trajectories from history-integral and grid-based calculations.** Panels (a,c) show trajectories computed with the present volume-grid-free implementation, while panels (b,d) show grid-based trajectories for the same reduced point-source model from Ref.¹⁰. Panels (a,b) use $Pe_{in} = 40$ and $\Lambda_t = 4$, and panels (c,d) use $Pe_{in} = 60$ and $\Lambda_t = 8$. Green and red markers denote the initial and final positions. The comparison concerns trajectory morphology; it does not report pointwise errors or fitted orbital parameters.

produce curling paths with comparable orbital scales in the two parameter cases. This agreement is a qualitative cross-check of trajectory morphology.

The panels display the two calculations separately and do not provide a grid-converged reference, a matched-time overlay, or a phase-error measure. They therefore cannot determine an error in the orbit radius or frequency, nor can they locate the onset of curling. Those questions require independently refined solutions and time-resolved trajectory differences.

B. Diffusion-dominated patterned surfaces

We next calculate leading-order drift velocities for two circular surface patterns. These calculations test the geometry and the permeable-interface convention of the model. They are not benchmarks for an impermeable, three-dimensional hydrodynamic swimmer.

For a two-slice pattern, let the active cap occupy $-\theta_c \leq \theta \leq \theta_c$ and set

$$\sigma(\theta) = \begin{cases} \sigma_0, & |\theta| \leq \theta_c, \\ 0, & \theta_c < |\theta| \leq \pi, \end{cases} \quad \Lambda_t(\theta) = \Lambda_0. \quad (36)$$

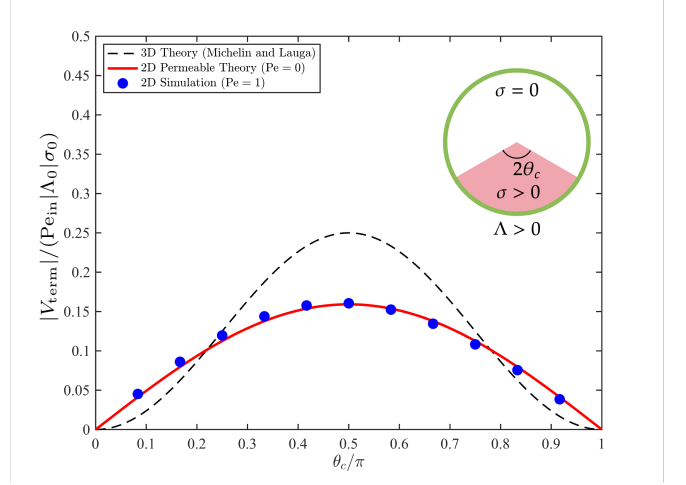


FIG. 3. **Dependence of the terminal speed on active-cap angle.** The dashed black curve shows the three-dimensional impermeable-sphere result of Michelin and Lauga.¹² The red curve is the two-dimensional permeable prediction in Eq. (39), and the blue markers are the reported two-dimensional calculations at the nominal input value $Pe_{in} = 1$. The ordinate is $|V_{term}|/(Pe_{in}|\Lambda_0|\sigma_0)$. The finite-input markers reproduce the symmetry and the single maximum but do not constitute a $Pe_{in} \rightarrow 0$ convergence test.

The nonconstant Fourier coefficients of the activity are

$$a_n = \frac{2\sigma_0}{n\pi} \sin(n\theta_c), \quad n \geq 1. \quad (37)$$

Solving Eq. (19) inside and outside the circle, with continuity of the field and the prescribed derivative jump, gives the mean-free surface field

$$\bar{C}_{mf}(1, \theta) = \sum_{n=1}^{\infty} \frac{a_n}{2n} \cos(n\theta). \quad (38)$$

The factor 1/2 follows because the source diffuses on both sides of the nominal interface. With uniform response, orthogonality leaves only the $n = 1$ mode in the translational drift. The leading speed is

$$V_0(\theta_c) = \frac{Pe_{in}|\Lambda_0|\sigma_0}{2\pi} \sin \theta_c + o(Pe_{in}). \quad (39)$$

Thus the two-dimensional permeable model predicts a single maximum at $\theta_c = \pi/2$. The corresponding three-dimensional impermeable problem has different geometry and boundary conditions and should not be used as a quantitative validation of Eq. (39).¹²

For the three-slice pattern, a source cap at the front and a sink cap at the rear are separated by inactive belts:

$$\sigma(\theta) = \begin{cases} \sigma_0, & |\theta| \leq \theta_0, \\ -\sigma_0, & |\theta - \pi| \leq \theta_0 \pmod{2\pi}, \\ 0, & \text{otherwise.} \end{cases} \quad (40)$$

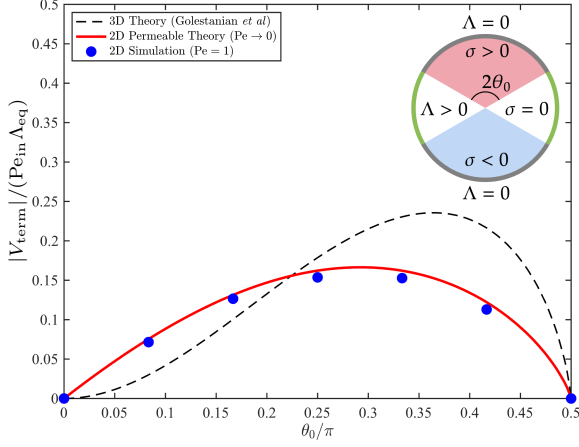


FIG. 4. **Terminal speed for the three-slice pattern.** The dashed black curve shows the three-dimensional impermeable-particle result of Golestanian *et al.*,¹¹ the red curve shows the two-dimensional permeable prediction from Eq. (43), and the blue markers are the reported calculations at the nominal input value $Pe_{in} = 1$. The ordinate is $|V_{term}|/(Pe_{in}\Lambda_{eq})$. The two theoretical curves differ in both dimension and permeability, and the finite-input markers provide only a qualitative comparison with the diffusion-dominated curve.

Only the inactive belts respond, with $\Lambda_t = \Lambda_{eq}$ there and $\Lambda_t = 0$ on the two active caps. The activity has the expansion

$$\sigma(\theta) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \sigma_n \cos(n\theta), \quad \sigma_n = \frac{4\sigma_0}{n\pi} \sin(n\theta_0), \quad (41)$$

and hence

$$\bar{C}(1, \theta) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{\sigma_n}{2n} \cos(n\theta). \quad (42)$$

The corresponding leading speed is

$$V_0 = \frac{Pe_{in}}{2\pi} \left| \int_0^{2\pi} \Lambda_t(\theta) \frac{\partial \bar{C}}{\partial \theta} \sin \theta d\theta \right| + o(Pe_{in}). \quad (43)$$

This expression provides a reference for the present permeable circular model. Comparison with the three-dimensional impermeable construction of Golestanian *et al.* changes both dimension and boundary condition, so it does not isolate either effect.¹¹

C. Wake-memory trajectories of patterned particles

We now set $\alpha = 0$ and $\Omega_0 = 0$ and prescribe an anti-symmetric active head. In body-fixed polar coordinates,

$$\sigma(\theta) = \begin{cases} \sigma_0, & 0 \leq \theta \leq \theta_c, \\ -\sigma_0, & -\theta_c \leq \theta < 0, \\ 0, & \text{otherwise,} \end{cases} \quad \sigma_0 = 1. \quad (44)$$

The translational and rotational response patterns are negative on the active head and positive on the inert remainder,

$$\Lambda_t(\theta) = \Lambda_r(\theta) = \Lambda(\theta) = \begin{cases} -0.5, & |\theta| \leq \theta_c, \\ 0.5, & \theta_c < |\theta| \leq \pi. \end{cases} \quad (45)$$

Configuration 1 has $\theta_c = \pi/4$, while Configuration 2 has $\theta_c = \pi/2$.

Figure 5 shows the trajectories through $t = 1000$ for four nominal input values. At the lowest value, both configurations remain in localized looping paths over the displayed interval. Several higher-input cases occupy a larger region and contain alternating curved and more extended segments. The sampled cases show a change of trajectory morphology, but they do not locate a critical parameter or establish a bifurcation. The label Pe_{eff} in the panels denotes the reported output swimming Péclet number, rather than the input coefficient Pe_{in} .

For a circular particle, the rotational closure reduces to

$$\Omega_{c,z}(t) = -\frac{1}{2\pi} \int_{-\pi}^{\pi} \Lambda_r(\theta) \frac{\partial C(1, \theta, t)}{\partial \theta} d\theta. \quad (46)$$

Evaluating this integral for the computed startup fields gives a positive initial angular velocity for both configurations. This is a statement about the initial field, not the sign of $\Omega_{c,z}$ throughout the trajectory. At later times, the particle repeatedly approaches regions of concentration deposited during earlier motion. The overlays in Fig. 5 associate the expanded paths with these wake encounters. They do not separate the head and tail contributions or identify a causal delay. The shooting-and-turning description is therefore a working interpretation of the computed trajectories.

V. DISCUSSION

The numerical extension in this work is the history-local evaluation of free-space diffusion generated by a moving surface distribution. The point-source Green-function history integral was already used in Ref.¹⁰. Here it is extended to rotating surface sources and surface-dependent translational and rotational response. The distinction matters for the claim of novelty: the present contribution is not the heat-kernel representation by itself.

Removing the volume grid changes which degrees of freedom are resolved, but it does not by itself make the calculation fast. Direct surface-source summation scales as $O(N_t^2 N_s^2)$ over a trajectory of N_t steps, and the regularized diagonal interaction introduces the additional parameters M and ϵ . No matched-error timing data are available in the present study. The method should therefore be described as volume-grid-free rather than computationally accelerated.

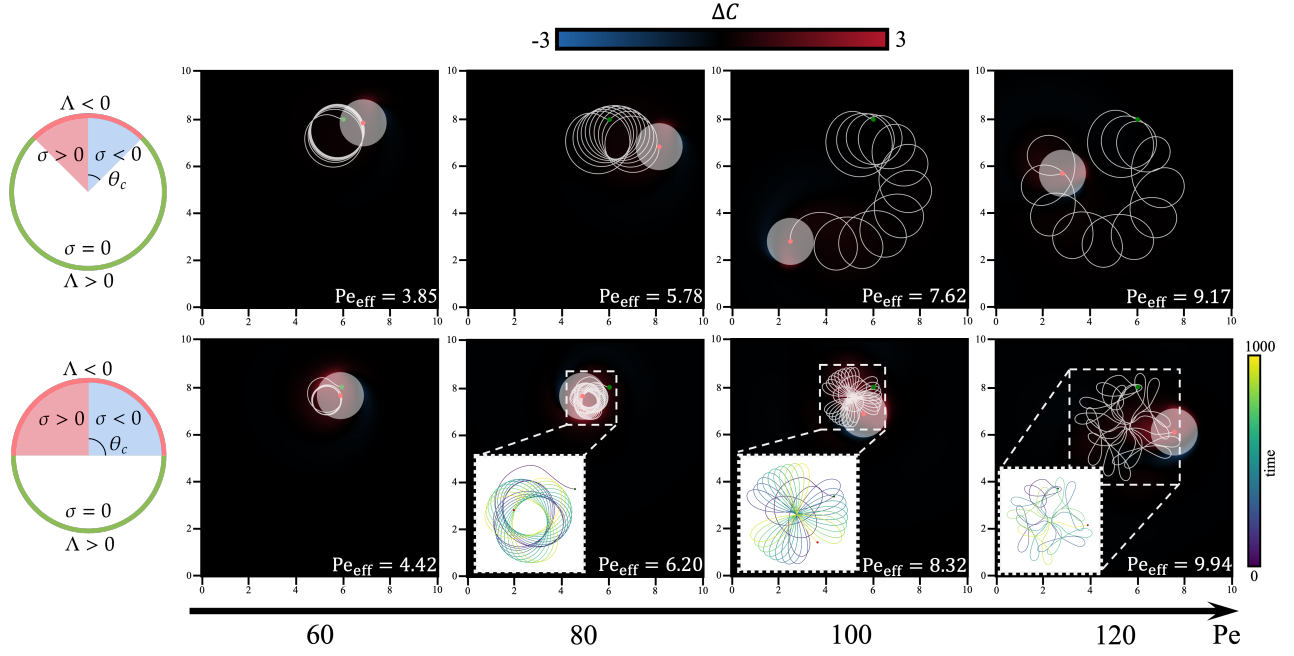


FIG. 5. **Representative trajectories and concentration fields for the two patterned particles.** The upper and lower rows show Configurations 1 ($\theta_c = \pi/4$) and 2 ($\theta_c = \pi/2$), respectively. From left to right, the trajectory panels use the nominal input values $Pe_{in} = 60, 80, 100$, and 120 . The reported effective values printed in the panels are $3.85, 5.78, 7.62$, and 9.17 for Configuration 1 and $4.42, 6.20, 8.32$, and 9.94 for Configuration 2. White curves show trajectories through $t = 1000$ superposed on the signed concentration deviation. In the surface schematics, the red and blue sectors have $\sigma = 1$ and -1 with $\Lambda = -0.5$, while the remaining surface has $\sigma = 0$ and $\Lambda = 0.5$. The panels show representative morphologies; they do not define a transition threshold.

The existing checks answer limited questions. The isotropic comparison in Fig. 2 tests whether two implementations of the same reduced point-source model produce similar path morphologies. The Fourier calculations in Sec. IV B fix the leading dependence on surface geometry for a permeable circular source. Neither check validates ambient-flow effects. A convergence claim for the surface-source algorithm also remains open until Δt , N_s , M , and ϵ are refined independently against defined reference quantities.

Within these limits, the patterned-particle calculations show that delayed diffusion is sufficient in the reduced model to produce localized loops and more extended, slingshot-like paths. The instantaneous angular velocity is a local surface integral, whereas the curvature and handedness of a long trajectory depend on the accumulated history. This distinction offers a useful way to interpret cases in which short-time rotation and large-scale path geometry differ. The current parameter samples and field overlays do not establish a phase transition or a unique microscopic mechanism.

The model has four physical restrictions. First, the scalar diffuses through the nominal particle interior, so the surface represents a permeable source rather than an impermeable solid. Second, the ambient fluid velocity, fluid-mediated solute advection, and the hydrodynamic conversion of slip into rigid-body motion are ab-

sent. Third, the activity and response are prescribed step functions rather than the outcome of reaction kinetics. Fourth, thermal noise, external boundaries, and particle deformation are omitted. These restrictions preclude a quantitative comparison with fully coupled autophoretic calculations or experiments without additional tests.

A direct test of the proposed wake-memory interpretation would resolve the translational and rotational contributions from each surface sector over one complete trajectory cycle. Turning off the inert-tail response or truncating the stored history would then provide falsification tests. A comparison with a coupled Stokes–solute model would determine whether the same trajectory classes persist when impermeability and hydrodynamic flow are restored.

VI. CONCLUSION

We have formulated a volume-grid-free history integral for a reduced autochemotactic particle with surface-distributed activity and response. A history–local split treats the most recent part of the diffusion integral, and a regularized inverse-time quadrature handles the near-diagonal surface interaction. The formulation extends an earlier point-source calculation to rotating surface patterns. For the reported cases, it reproduces the morphol-

ogy of selected grid-based point-source trajectories and generates localized and slingshot-like paths for two patterned particles. The associated fields are consistent with delayed encounters with a deposited chemical wake.

The present evidence does not establish a computational speed advantage, a universal convergence order, a dynamical transition, or equivalence to a fully coupled hydrodynamic swimmer. Independent refinement tests, matched-error timing measurements, and mechanism-specific ablations are the next requirements for those claims.

DATA AVAILABILITY

The current manuscript package contains the text and plotted figures but does not contain the numerical source data or simulation code. A citable archive containing the data, parameter files, and scripts required to reproduce the figures should be supplied before submission.

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